Cross-Event Optical Relation Distillation for SAR Landslide Mapping

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***Abstract*—Optical imagery maps landslides most accurately, but clouds often leave synthetic aperture radar (SAR) as the only rapid observation after a disaster. We use optical data only during training to improve a landslide mapper that receives SAR backscatter and terrain at inference. Landslide inventories are collected event by event, and a SAR network trained on them learns the training events rather than landslides; distilling the outputs or features of an optical teacher does not resolve this, because the teacher reproduces the training labels and most optical content cannot be observed in SAR. We find that students generalize better when they do not copy the teacher's pairwise region relations. We therefore distill only the distribution of each region's similarities to regions of other disaster events, using a closed-form one-dimensional Wasserstein distance, and add to the student decoder one bank of prototypes shared by all events and three decoder scales. On the official split of a 23-event Sentinel-1/2 benchmark, the student reaches 56.45% intersection over union (IoU) and 77.59% average precision (AP) with SAR and terrain input, compared with 52.20% and 72.91% for the same U-Net trained without optical data.**

***Index Terms*—Knowledge distillation, landslide mapping, privileged information, synthetic aperture radar (SAR), Sentinel-2.**

I. Introduction

SAR supports landslide mapping when clouds or darkness prevent optical acquisitions during emergency response [1]–[3]. Convolutional networks detect landslides from SAR backscatter [4]–[6], yet a recent multi-sensor benchmark shows that optical features remain clearly more accurate than SAR features, and that SAR models degrade most on events not represented in training [7].

Training archives frequently contain co-registered optical imagery even when an emergency map must rely on SAR. Learning with privileged information and knowledge distillation (KD) exploit such training-only inputs [8], [9]. Remote-sensing KD transfers predictions, features, or attention between modalities [10]–[13], and relational KD transfers similarities between samples [14]–[16].

Two properties of landslide data limit these approaches. First, inventories are grouped by triggering event. A SAR U-Net reaches 90.9% IoU on validation patches from its training events but 52.2% on the test events, so it largely memorizes event-specific appearance. Second, a multimodal teacher fits the training labels almost exactly (96.1% training IoU), so its outputs add little beyond the labels, while its features encode optical evidence that SAR cannot reproduce. We further observe that students that reproduce the teacher's pairwise region relations more closely generalize worse to new events.

We therefore transfer the teacher's relational structure rather than its content, and only the part that is not tied to individual training events. Our contributions are: 1) a cross-event relation distillation that matches, for each region, the distribution of its similarities to regions of other events with a closed-form Wasserstein distance and no pair correspondence; 2) a prototype bank shared by all events and by three decoder scales, which gives this relational structure a deployable form for landslides of very different sizes; 3) an analysis on a 23-event benchmark showing why output, feature, and pairwise relation transfer underperform for event-grouped landslide data.

II. Method

A. Problem Formulation

Let X_s denote the student input: pre- and post-event Sentinel-1 γ^0 VV/VH backscatter, elevation, and slope (six channels). The teacher additionally receives pre- and post-event Sentinel-2 bands X_o (twelve channels). Both networks follow the U-Net of [7], [17], with four decoder stages at 16, 32, 64, and 128 pixels. The teacher is trained with the segmentation loss and then frozen; the student is trained with the teacher and deployed on X_s only. Fig. 1 summarizes the method.

B. Scale- and Event-Shared Prototypes

After decoder stage $l \in \{2,3,4\}$ (32, 64, and 128 pixels), the student feature $f_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ is compared with one bank of K learnable prototypes $\{P_k\}$ shared by all events and stages:

$$a_{lk}(p) = \mathrm{softmax}_k\Big(\cos\big(W_q^l f_l(p), P_k\big)/\tau\Big), \tag{1}$$

$$f_l'(p) = f_l(p) + W_o^l \sum_k a_{lk}(p)\, W_v^l P_k, \tag{2}$$

where p is a pixel, W_q^l, W_v^l, and W_o^l are stage-specific 1×1 projections, $\tau = 0.1$, and $K = 16$. W_o^l is zero-initialized, so training starts from the plain U-Net. Every pixel of every event reads the same prototypes at every scale; a concept that the relation loss organizes at one scale is therefore available to large landslide bodies, boundaries, and small slides at the others. The prototypes remain in the deployed network.

C. Cross-Event Wasserstein Relation Distillation

The relation loss acts at stage 2 (32×32 pixels). Teacher and refined student features there are average-pooled over valid pixels into 4×4 regions per patch, each covering 320×320 m, and normalized to unit length, giving t_i (teacher) and s_i (student, from f_2'). For region i from event e_i, let $\mathcal{N}_i = \{j : e_j \neq e_i\}$

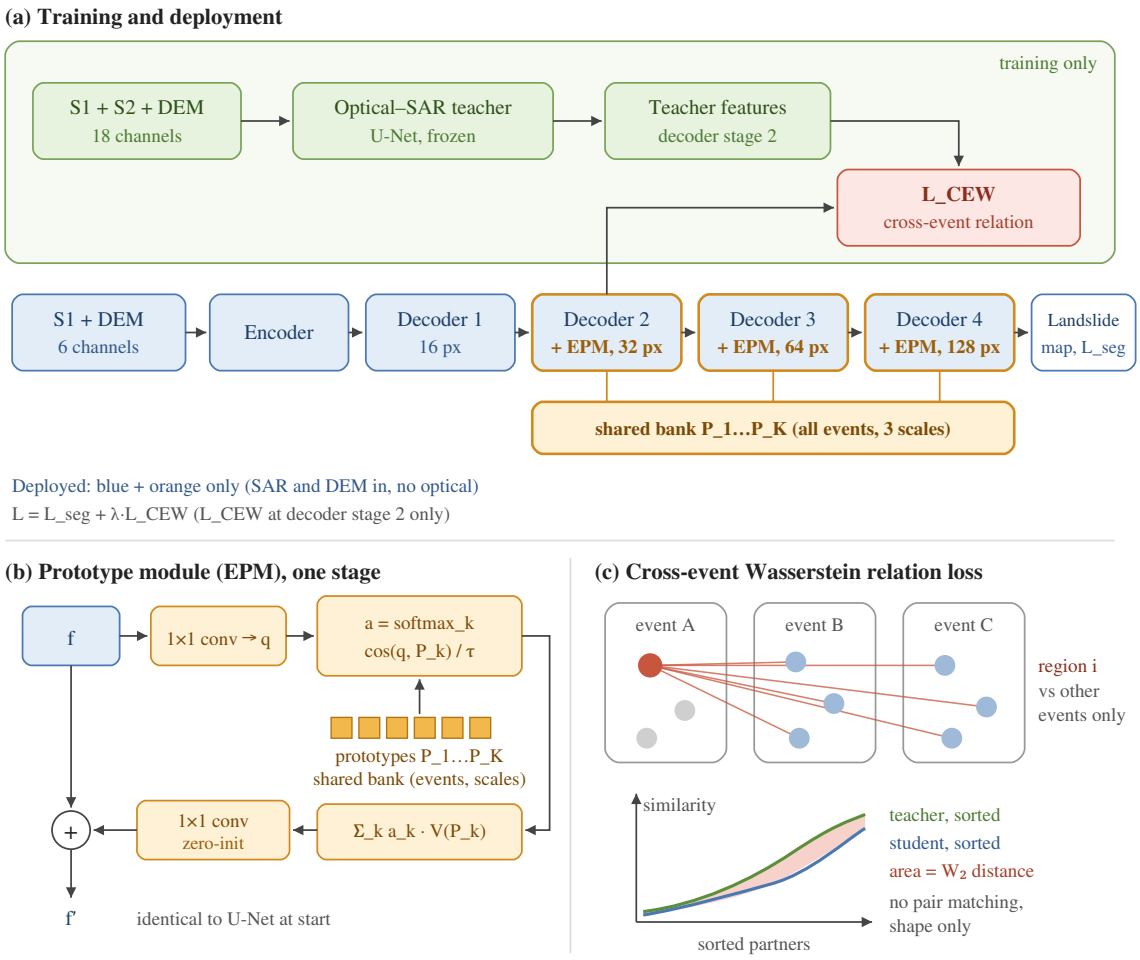


Fig. 1. (a) Training and deployment. The optical–SAR teacher (green) and L_{CEW} are used only during training; the deployed student (blue and orange) receives SAR and terrain. (b) Prototype module (EPM) at one decoder stage. (c) Cross-event Wasserstein relation loss: each region is compared only with regions of other events; teacher and student similarities are sorted separately and compared as distributions.

be the regions of other events in the mini-batch. The teacher and student similarity sets are

$$\mathscr{T}_i = \{t_i^\top t_j\}_{j\in\mathscr{N}_i}, \qquad \mathscr{S}_i = \{s_i^\top s_j\}_{j\in\mathscr{N}_i}, \tag{3}$$

each shifted by its minimum over all cross-event pairs in the batch. Sorting each set separately gives $\tilde{\mathscr{T}}_i$ and $\tilde{\mathscr{S}}_i$, and the squared one-dimensional Wasserstein distance between the two empirical distributions is

$$L_{\mathrm{CEW}} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{|\mathscr{N}_i|}\sum_{r=1}^{|\mathscr{N}_i|}\left(\tilde{\mathscr{T}}_i[r]-\tilde{\mathscr{S}}_i[r]\right)^2. \tag{4}$$

Sorting removes pair correspondence: the student decides which regions of other events are close or distant, while the teacher prescribes how the cross-event similarities are distributed. Same-event pairs never enter the loss, so acquisition conditions shared within an event cannot satisfy it, and no partner count or quantile number has to be chosen.

The loss relates whole regions across events, which requires features that describe the content of a region rather than its texture. At stage 2, each region is summarized from 8×8 cells of a deep decoder layer. The 64- and 128-pixel stages are not constrained; they read the same prototypes, so the organization learned at stage 2 reaches boundaries and small slides without imposing the teacher's fine-scale optical texture on SAR features. Section III-E compares all four stages.

D. Learning Objective

The student minimizes

$$L = L_{\mathrm{BCE}} + L_{\mathrm{Dice}} + \lambda L_{\mathrm{CEW}}, \tag{5}$$

with binary cross-entropy and foreground soft Dice on valid pixels. We set $\lambda = 30$, $K = 16$, and $\tau = 0.1$ before testing; at initialization λL_{CEW} then contributes about 20 times the gradient norm of the segmentation loss. The teacher and L_{CEW} are removed after training.

III. Experiments

A. Dataset and Settings

We use the Sentinel-1/2 landslide reference data of [7] with its official split: 355 training and 88 validation patches from 23 events, and 120 test patches of 128×128 pixels at 10 m. The test set contains 44 patches from four independent events (Ind.) and 76 patches from four dependent events whose

TABLE I
TEST RESULTS (%). ROWS ABOVE THE REFERENCE BLOCK RECEIVE SAR AND TERRAIN AT INFERENCE. DISTILLATION METHODS SHARE THE U-NET STUDENT, INITIALIZATION, AND FROZEN OPTICAL–SAR TEACHER. BOLD: BEST SAR-INPUT VALUE.

Method	AP	IoU	F1	IoU Ind.	IoU Dep.
Direct segmentation					
U-Net [7]	72.91	52.20	68.59	63.24	46.03
UNet++ [18]	–	–	–	–	–
DeepLabv3 [19]	–	–	–	–	–
Distillation from the optical–SAR teacher					
Vanilla KD [9]	70.89	53.45	69.66	68.52	44.97
AT [13]	76.63	53.12	69.39	61.22	48.63
Multi-to-Uni [11]	74.92	53.28	69.52	63.14	47.88
ACAM-KD [12]	74.54	54.59	70.63	66.09	48.45
PKDOT [16]	74.75	55.74	71.58	**70.52**	47.80
DIS2 [10]	75.85	55.97	71.77	69.71	48.06
Ours	**77.59**	**56.45**	**72.17**	69.46	**49.88**
Optical input at inference (reference only)					
Optical U-Net	75.90	66.17	79.64	84.71	58.38
Optical–SAR U-Net (teacher)	85.09	70.38	82.62	81.90	64.88

training patches are separated by a spatial block split (Dep.). Inputs follow the one-pre/one-post setting of [7]: γ^0 VV/VH in decibels, Copernicus DEM elevation, and slope; the teacher adds Sentinel-2 B2, B3, B4, B8, B11, and B12 from the pre- and post-event scenes.

All U-Nets, including the teacher and every student, use AdamW with a learning rate of 10^{-4}, weight decay of 0.01, gradient clipping at 1, batches of 16, random flips and 90° rotations, and 200 epochs (constant rate for 100 epochs, then cosine decay to 1%). Validation IoU, computed every 20 epochs, selects the checkpoint; each model is tested once after selection, at threshold 0.5. All students share one initialization, one data order, and the same frozen teacher. We report AP and pixel-pooled IoU and F1 over the 120 test patches, IoU on Ind. and Dep., and per-event F1 as in [7]. Each configuration is one training run with seed 42.

B. Main Comparison

Table I compares direct segmentation networks trained on the same input and recipe—the U-Net of [7] and UNet++ and DeepLabv3 with ImageNet ResNet-34 encoders [20]—with distillation methods that transfer outputs (Vanilla KD), attention or features (AT, Multi-to-Uni, ACAM-KD, DIS2), and pairwise relations (PKDOT). All distillation methods use our U-Net student and teacher. Ours improves IoU by 4.25 points and AP by 4.68 points over the U-Net, and gives the highest pooled AP, IoU, and F1 among all SAR-input models, 0.48 IoU points above DIS2 and 0.96 AP points above the best competing AP. The gain comes mainly from recall (70.3% versus 58.4%), at lower precision (74.2% versus 83.1%). On the independent events, PKDOT and DIS2 remain 1.06 and 0.25 points higher; Ours is highest on the dependent events. Vanilla KD gives the lowest AP, consistent with a teacher that reproduces the training labels. Optical inputs at inference remain far more accurate, which bounds what training-only optical data can contribute.

Fig. 2 shows test patches with cloud-free optical scenes. On the rockslide in (a), Ours and PKDOT recover the margins that the U-Net misses. On the loess slides in (b), Ours recovers slopes that all other students miss, at the cost of additional false positives, similar to the teacher. In (c) all students produce similar maps. In (d), small peat slides are largely missed by all models, including the teacher.

TABLE II
TEST F1 (%) PER EVENT AND MEAN OVER THE FOUR INDEPENDENT (IND.) AND FOUR DEPENDENT (DEP.) EVENTS.

	CN	CO	IN	PH	Ind.	BR	CA	IE	KG	Dep.
U-Net	87	33	36	65	55.3	83	78	38	50	62.3
DIS2	92	36	44	68	59.7	84	79	48	52	65.8
Ours	91	30	39	77	59.0	81	76	20	60	59.2
Teacher	94	89	70	96	87.1	91	74	0	78	60.7

C. Per-Event Results

The dataset paper reports per-event F1 averaged over events rather than pooled scores; for γ^0 VV/VH with terrain it gives a mean F1 of 0.68 over the dependent events with its own training [7]. Under our recipe the U-Net reaches 62.3% (Table II). Ours raises F1 on the large Chinese rockslide (CN), the Philippine slides (PH), and the Kyrgyz loess slides (KG), but falls on the small Irish peat slides (IE), which the teacher misses entirely. Because Ours transfers the teacher's relational structure, it also inherits this blind spot. Averaged over events, Ours (59.1%) is close to the U-Net (58.8%) and below DIS2 (62.8%); its advantage lies in pixel-pooled scores, which are dominated by the larger landslides.

D. Why Pairwise Relations Transfer Poorly

For every training and validation region we take the 30 regions of other events that the teacher finds least similar and measure the share that a frozen student also ranks among its own 30 least similar. Across 13 students trained with the same recipe, this agreement correlates negatively with test IoU ($r = -0.70$; Fig. 3). The best students agree with the teacher less than the U-Net does, whereas losses that enforce pair correspondence raise the agreement to 0.68–0.82 while test IoU falls to 52–53%. Pairwise teacher relations therefore carry structure tied to the training events; the distribution of cross-event similarities is what transfers.

E. Ablation Study

Table III isolates each component. L_{CEW} alone raises IoU from 52.20% to 55.57%. The shared prototypes alone leave IoU unchanged (52.07%) but raise AP to 77.99%: they improve the ranking of landslide pixels without changing decisions at threshold 0.5. Combined with L_{CEW}, this ranking turns into detections, mainly through recall. Separate prototypes at 32 and 64 pixels with the loss at both stages lower IoU to 55.22% and independent-event IoU to 64.85%, whereas one bank shared by three stages and constrained only at 32 pixels gives the best result. [STAGE RESULT TO BE INSERTED.]

Losses that impose pair correspondence fall to 52.4–52.5% IoU. Pairing each region with the teacher's 30 least similar partners and matching them by optimal transport (OT), as

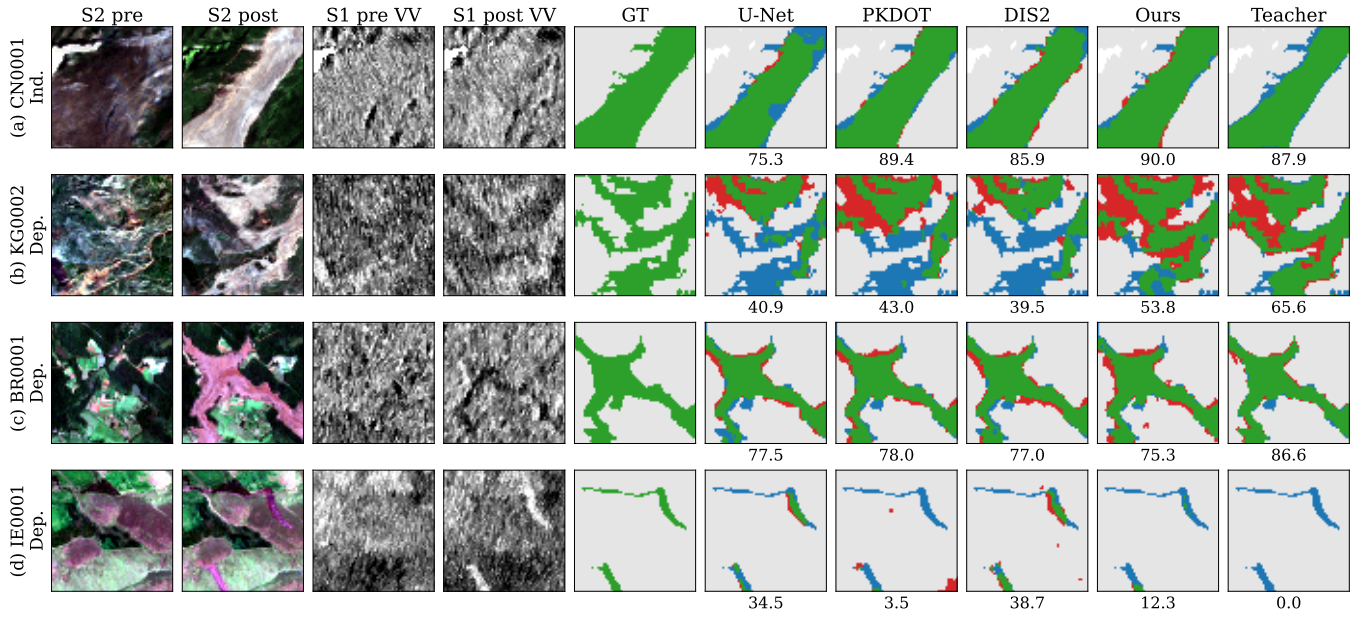


Fig. 2. Test patches with cloud-free pre- and post-event Sentinel-2 scenes. Optical (true-color) and SAR panels show inputs; only the teacher receives the optical scenes. GT: reference (green). Predictions at threshold 0.5: true positives green, false positives red, false negatives blue; white marks invalid SAR pixels. Numbers give patch IoU (%). (a), (b) cases where Ours changes the map; (c) all students are similar; (d) small peat slides that the teacher also misses.

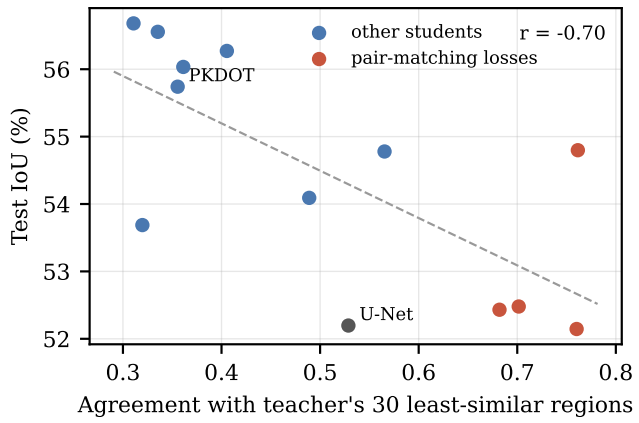


Fig. 3. Teacher–student agreement on the teacher's 30 least similar cross-event regions (validation patches) versus test IoU for students trained with the same recipe.

in PKDOT, and summarizing the distribution by 32 quantiles matched by OT reach scores similar to the Wasserstein distance (56.68% and 56.55%); the former repeated at 56.03%. The Wasserstein form needs no partner count, quantile number, or iterative solver. Label-defined relations trail optical-teacher relations by 1.65 points, so the optical teacher contributes beyond the labels.

[SENSITIVITY TEXT TO BE INSERTED.]

F. Deployment Cost

The three prototype modules add 0.14 M parameters and 0.36 G MACs to the U-Net (about 3%, Table IV). The other distillation students deploy the unchanged U-Net. The teacher and L_{CEW} are used only in training, so inference uses SAR and terrain alone.

TABLE III
ABLATION (%). ALL ROWS USE THE SAME U-NET, INITIALIZATION, TEACHER, AND RECIPE. †REPEATING THIS RUN WITH IDENTICAL SETTINGS GAVE 56.03% IOU.

Configuration	AP	IoU	IoU Ind.
Components			
U-Net	72.91	52.20	63.24
+ shared prototypes (no optical)	77.99	52.07	64.54
+ L_{CEW} (no prototypes)	76.41	55.57	68.85
+ prototypes at 32 px + L_{CEW}	77.58	56.10	70.84
+ stage-wise prototypes, L_{CEW} at 32/64 px	75.51	55.22	64.85
+ shared prototypes + L_{CEW} (Ours)	77.59	56.45	69.46
Stage of L_{CEW} (shared prototypes)			
16 px	–	–	–
32 px (Ours)	77.59	56.45	69.46
64 px	–	–	–
128 px	–	–	–
32, 64, and 128 px	–	–	–
Relation target (prototypes at 32 px)			
Pairs, rank-stratified partners	72.55	52.43	66.83
Pairs, soft correspondence (KL)	72.43	52.48	65.66
Pairs, 30 least similar, OT†	76.83	56.68	69.01
Distribution, 32 quantiles, OT	78.59	56.55	68.77
Distribution, Wasserstein	77.58	56.10	70.84
Relation source (PKDOT loss, no prototypes)			
Label-defined relations	74.96	54.09	65.78
Optical-teacher relations	74.75	55.74	70.52
Optical, cross-event pairs only	77.24	56.27	71.29

IV. Conclusion

For event-grouped landslide inventories, an optical teacher transfers best to a SAR student through the distribution of its cross-event relations rather than through outputs, features, or pairwise relations. Matching this distribution with a closed-form Wasserstein loss and reading it through prototypes shared across events and scales improves SAR-only landslide mapping at about 3% additional deployment cost. The method

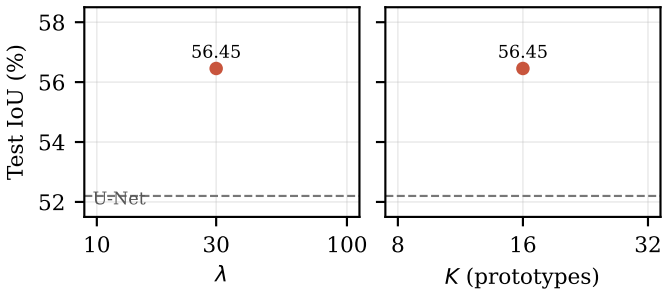


Fig. 4. Test IoU of Ours for the relation-loss weight λ (left, $K = 16$) and the number of prototypes K (right, $\lambda = 30$). Dashed line: U-Net. Values were fixed at $\lambda = 30$, $K = 16$ before testing.

TABLE IV
DEPLOYED PARAMETERS AND MULTIPLY–ACCUMULATE OPERATIONS (MACS) FOR ONE 128×128 PATCH.

Network	Params (M)	MACs (G)
U-Net and KD students	31.05	12.07
UNet++ (ResNet-34)	26.09	4.63
DeepLabv3 (ResNet-34)	26.02	6.86
Ours	31.19	12.43
Teacher (training only)	31.05	12.18

raises recall at some cost in precision, leads on pixel-pooled rather than event-averaged scores, and inherits the teacher's failures on small landslides that optical data also miss.

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